Analysis and Design of Wireless Power Transfer Systems for Biomedical Implants

ECE 366/566: Neural Engineering & Implantable Devices

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ABSTRACT

This paper presents a simulation and experimental investigation of inductively coupled wireless power transfer (WPT) systems designed for biomedical implant applications. Two-coil and three-coil resonant and non-resonant configurations are modelled in LTspice and evaluated across the 2–8 MHz frequency range, with a primary operating frequency of 5 MHz. Power transfer efficiency (PTE) is analytically derived from simulated current waveforms and validated against S-parameter measurements obtained using a vector network analyzer (VNA). Resonant tuning with series capacitors is shown to improve PTE by 1.6× in the two-coil system, and by approximately 295× in the three-coil system compared with their non-resonant counterparts. A parametric study of PTE versus coil separation (0–40 mm) demonstrates that the three-coil resonant topology delivers PTEs that are consistently ~74× higher than the two-coil resonant case. VNA-measured S21 data confirm the qualitative trend of decreasing power transfer with increasing coil gap; however, the absolute measured PTEs exceed simulation values by 30–150×, attributable to model simplifications, parasitic effects, and inherent differences between S-parameter measurements and circuit-based PTE definitions.

I. INTRODUCTION

Biomedical implants such as cochlear stimulators, retinal prostheses, deep-brain stimulators, and cardiac rhythm management devices have fundamentally transformed modern medicine. A critical design challenge common to all such systems is reliable, long-term power delivery to deeply embedded electronics. Conventional battery-based approaches impose severe constraints on device volume, operational lifetime, and—crucially—patient safety, since battery replacement requires invasive surgical procedures. Wireless power transfer offers a compelling alternative: energy is delivered transcutaneously via time-varying magnetic fields, eliminating both the battery and its associated surgical risk.

Inductively coupled WPT systems exploit Faraday's law of electromagnetic induction to transfer power between spatially separated coils. At moderate frequencies (1–10 MHz), the technique is well-established, yet efficiency degrades rapidly with coil misalignment and separation—a particularly acute concern in implant scenarios where the transmitter sits external to the body and the receiver resides centimetres beneath the skin. Resonant operation, achieved by tuning each coil with a series or parallel capacitor to resonate at the drive frequency, markedly improves the effective coupling and therefore the PTE.

The canonical two-coil WPT architecture is simple but limited: PTE falls steeply when the coupling coefficient k is small. Three-coil topologies introduce an intermediate resonator on the implanted side, enabling a cascaded energy transfer that mitigates the sensitivity to weak primary coupling. The additional degree of design freedom—the strong coupling between the receiver and load coils on the implant—substantially boosts overall efficiency.

This work undertakes a systematic comparative study. Section II reviews the theoretical underpinning of inductive coupling and resonance. Section III describes the simulation methodology and experimental setup. Sections IV–VII present results for the two-coil, three-coil, parametric, and VNA validation experiments, respectively. Sections VIII–X provide engineering discussion, identified limitations, and conclusions.

II. THEORY AND BACKGROUND

A. Inductive Coupling and Mutual Inductance

When alternating current flows in a transmitter coil, a time-varying magnetic flux is generated. A fraction of that flux links with a nearby receiver coil, inducing a voltage according to Faraday's law. The extent of flux linkage is quantified by the mutual inductance M and, more conveniently, by the coupling coefficient:

where L1 and L2 are the self-inductances of the transmitter and receiver coils, respectively. The coefficient k lies in [0, 1]: unity corresponds to perfect coupling (lossless transformer), while values near zero characterise the loose coupling typical of implant geometries ($k \sim 0.01$ in this work for the transcutaneous link).

B. Resonant Frequency Tuning

Each coil possesses a self-inductance L and a parasitic series resistance R. Introducing a series capacitor C tunes the coil to a resonant frequency:

At resonance, the reactive impedance of the capacitor cancels that of the inductor, leaving only the small ohmic resistance. This impedance minimisation maximises current flow in the coil for a given drive voltage, substantially increasing the magnetic field amplitude and therefore the coupled power. Off-resonance, the impedance mismatch reduces current and degrades PTE.

C. Power Transfer Efficiency

The PTE is defined as the ratio of power delivered to the load to total input power consumed:

where the individual power terms are:

Here, I_Load, I_Source, and I_Coil,n are the RMS current magnitudes through the load, source, and nth coil resistance, respectively. These quantities are extracted from LTspice AC-sweep simulations.

D. Three-Coil Enhancement

In a three-coil system, the transmitter coil (coil 1) is loosely coupled to a receiver coil (coil 2) with $k12 \sim 0.01$, while coil 2 is tightly coupled to a load coil (coil 3) with $k23 \sim 0.3$ on the implanted side. Resonating all three coils creates a cascaded power path: the intermediate resonator (coil 2) amplifies the small voltage induced by the transcutaneous link and drives the load coil efficiently. The net result is that the tight k23 coupling partially compensates for the weak k12 coupling, yielding significantly higher PTE than the two-coil case.

III. METHODOLOGY

A. Coil Specifications

Two planar spiral coil types were used throughout, both wound with 20 AWG copper wire with 1 mm inter-turn spacing and 8 turns. Coil A (outer diameter 35 mm) was used for all receiver and load positions as well as the transmitter in the two-coil experiments. Coil B (outer diameter 70 mm) was used as the transmitter in the three-coil experiments. The derived coil inductances and resistances are: L_A = 1.25 μH, R_A = 0.018 Ω; L_B = 5.3 μH, R_B = 0.047 Ω.

B. LTspice Simulation Setup

All WPT circuits were modelled in LTspice XVII as series-coupled inductors with shared magnetic coupling (K directive). The AC sweep spanned 2 MHz to 8 MHz. The source and load resistances were 50 Ω, consistent with both the project specification and the VNA port impedance. Resonant capacitors were sized using equation (2): C1 = C2 = 816 pF for the two-coil system (tuned to 5 MHz with L = 1.25 μH), C1 = 190 pF for the three-coil transmitter (tuned with L = 5.3 μH), and C2 = C3 = 816 pF for the implanted coils. The AC source had an amplitude of 1 V.

Current magnitudes—mag(I(RSource)), mag(I(RLoad)), and mag(I(RCoil_n)) for each coil—were exported as ASCII text and post-processed in Python using equations (3)–(7) to compute PTE as a function of frequency. For the Part C parametric study, k12 was varied according to a measured distance–coupling lookup table, and LTspice .meas directives extracted currents at exactly 5 MHz for each distance step.

C. Experimental Setup (VNA)

Two-coil resonant experimental validation was performed using a two-port VNA. Coil A was connected to each port via calibrated SMA cables. The forward transmission S21 was measured at 5 MHz for coil separations of 10, 20, 30, and 40 mm. Experimental PTE was recovered from the measured S21 (in dB) via:

This conversion is exact for a 50 Ω matched system where S21 directly quantifies the power transfer ratio from port 1 to port 2.

IV. PART A — TWO-COIL WPT: RESONANT VS. NON-RESONANT

The two-coil system comprised a transmitter and receiver, each with L = 1.25 µH, R_coil = 0.018 Ω, and coupling coefficient k = 0.01. The non-resonant configuration had no capacitors; the resonant configuration added C1 = C2 = 816 pF to tune both coils to 5 MHz. Fig. 1 and Fig. 2 show the simulated PTE versus frequency for the non-resonant and resonant cases, respectively.

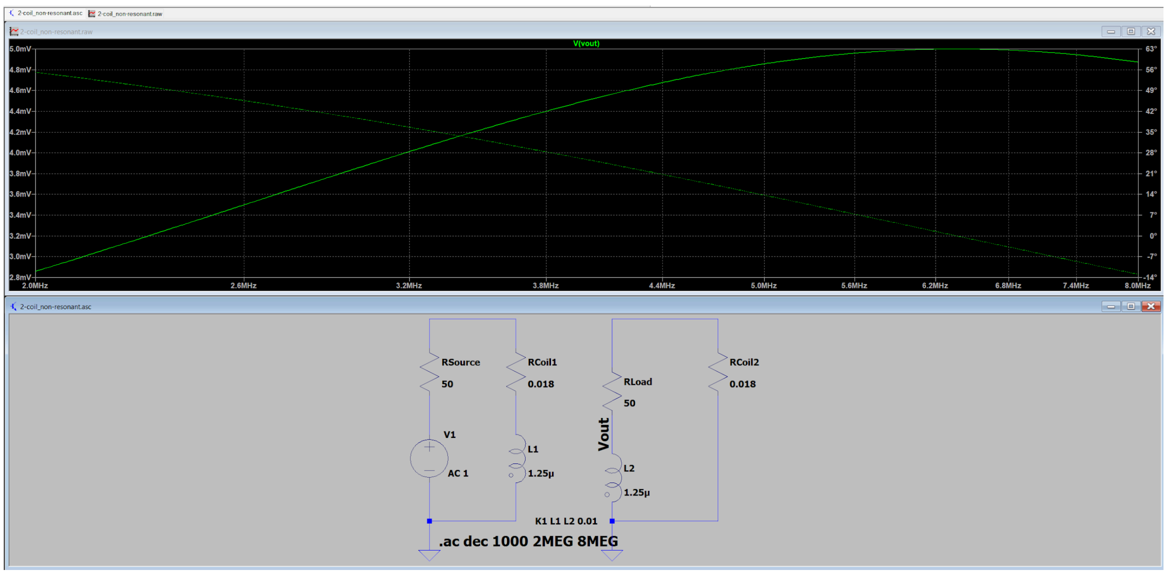

Fig. 1. Simulated PTE vs. frequency for the two-coil non-resonant WPT system (k = 0.01).

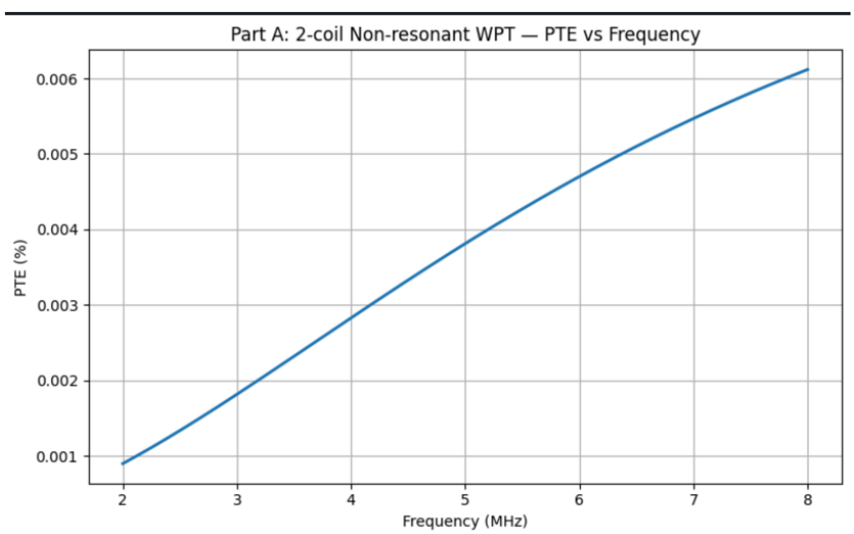

Fig2. Part A: 2-coil Non-resonant WPT – PTE vs Frequency

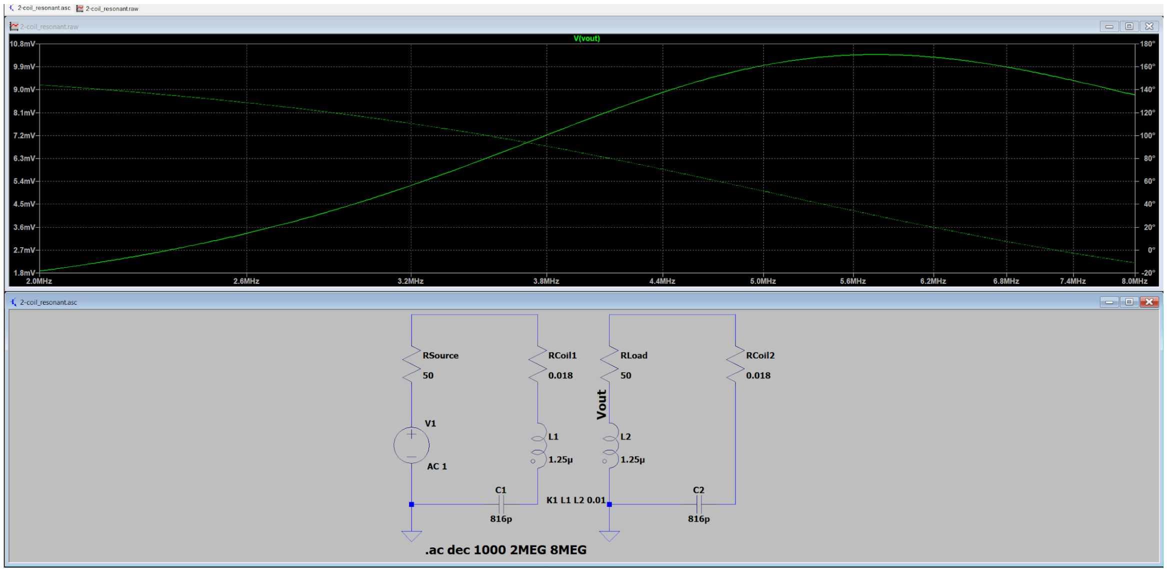

Fig. 3. Simulated PTE vs. frequency for the two-coil resonant WPT system (k = 0.01, C = 816 pF).

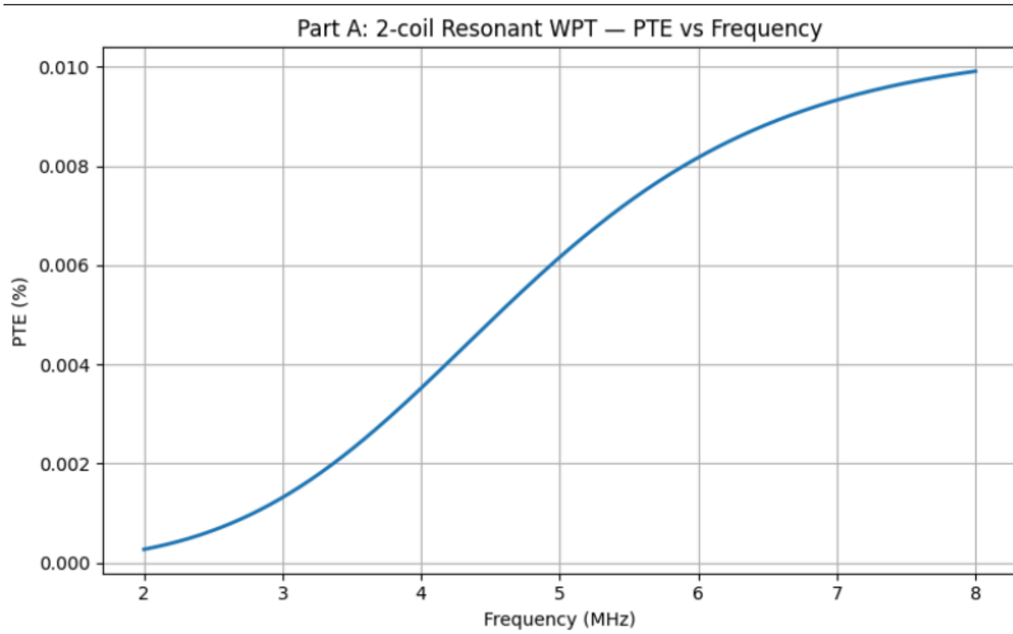

Fig. 4. 2- coil resonant WPT – PTE vs Frequency

At 5 MHz, the non-resonant system achieves PTE = 0.003812%, while the resonant system achieves PTE = 0.006163%—a 1.616× improvement. Table I summarises the key metrics.

Parameter	Non-Resonant	Resonant
PTE at 5 MHz	0.003812 %	0.006163 %
Max. PTE	0.006118 % @ 8 MHz	0.009914 % @ 8 MHz
P_Load at 5 MHz	4.715×10^{-7} W	1.232×10^{-6} W
V_Load at 5 MHz	4.855×10^{-3} V	7.848×10^{-3} V

Table I. Two-coil WPT simulation results at 5 MHz.

Although resonance improves PTE, the absolute values remain very low because k = 0.01 represents extremely weak coupling. The resonant PTE curve does not exhibit a sharp peak at 5 MHz; instead, PTE increases monotonically across the simulated band. This behaviour arises because PTE is a ratio of power terms whose frequency dependencies partially cancel. The load power has a broad maximum near 6.2 MHz, while the source loss—dominant in the denominator—also evolves with frequency, causing the PTE ratio to continue rising beyond 5 MHz.

V. PART B — THREE-COIL WPT: RESONANT VS. NON-RESONANT

The three-coil system used Coil B as the transmitter (L1 = 5.3 µH, R = 0.047 Ω) and two Coil A units as the receiver and load coils (L2 = L3 = 1.25 µH, R = 0.018 Ω each). Coupling coefficients were k12 = 0.01 (transcutaneous link) and k23 = 0.3 (implanted side). The resonant configuration employed C1 = 190 pF, C2 = C3 = 816 pF. Fig. 3 and Fig. 4 present the PTE versus frequency results.

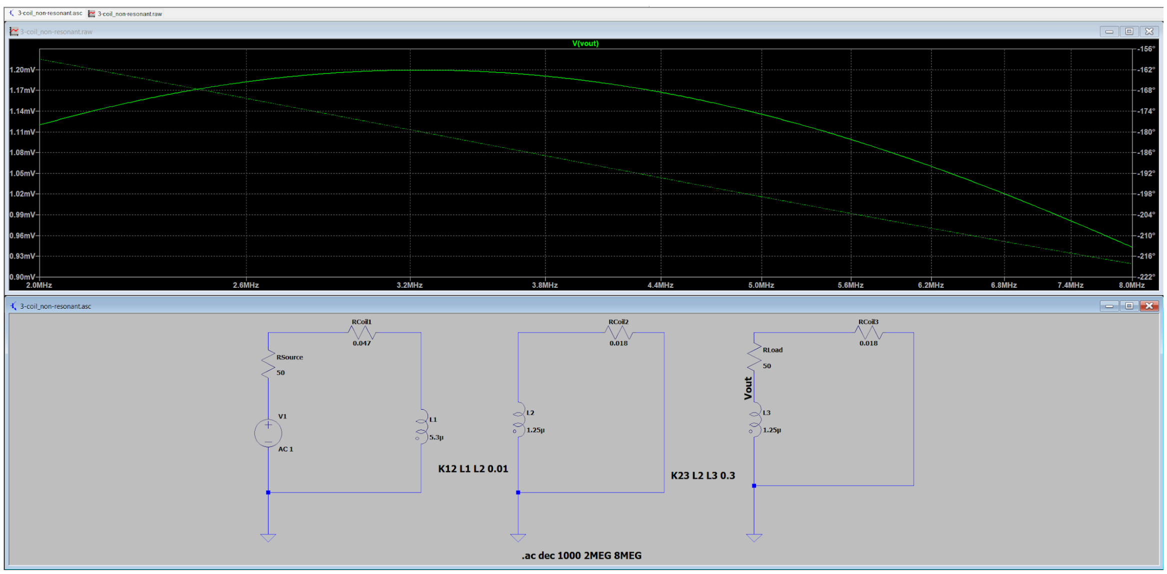

Fig. 5. Simulated PTE vs. frequency for the three-coil non-resonant WPT system.

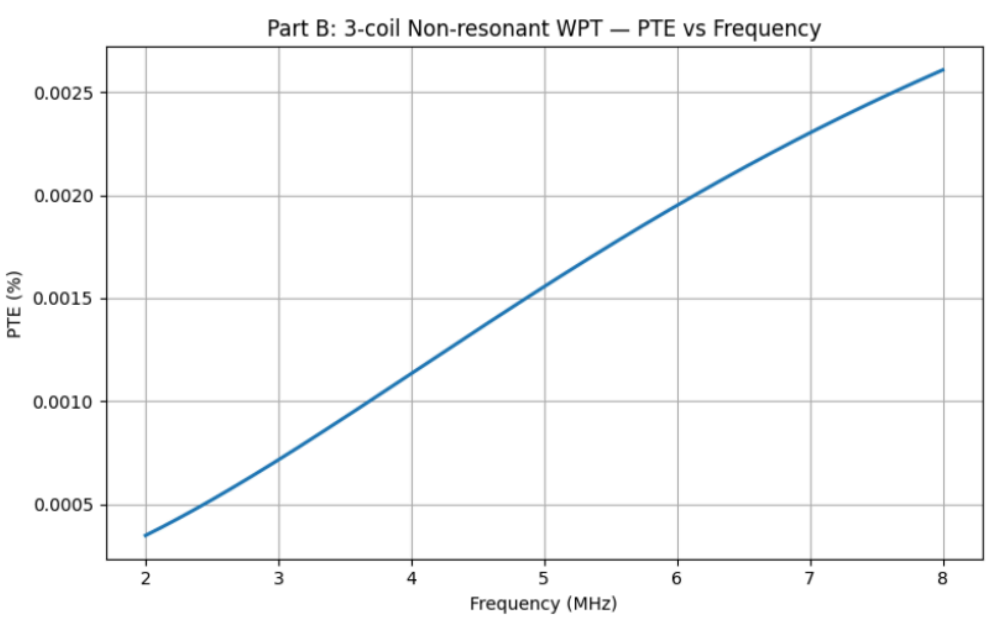

Fig. 6. 3- coil Non- resonant WPT – PTE vs Frequency

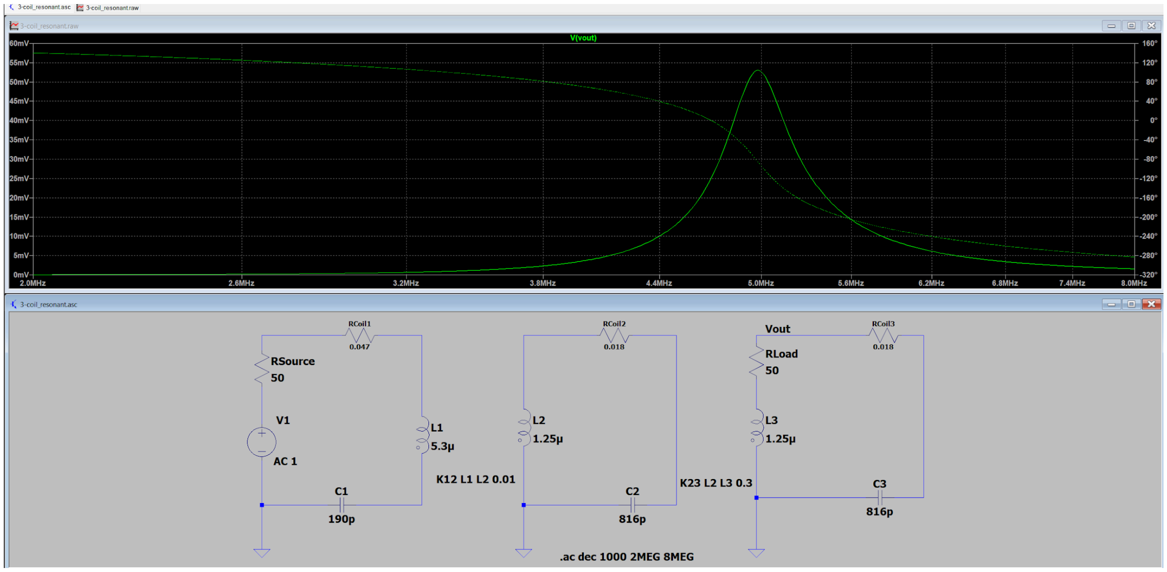

Fig. 7. Simulated PTE vs. frequency for the three-coil resonant WPT system.

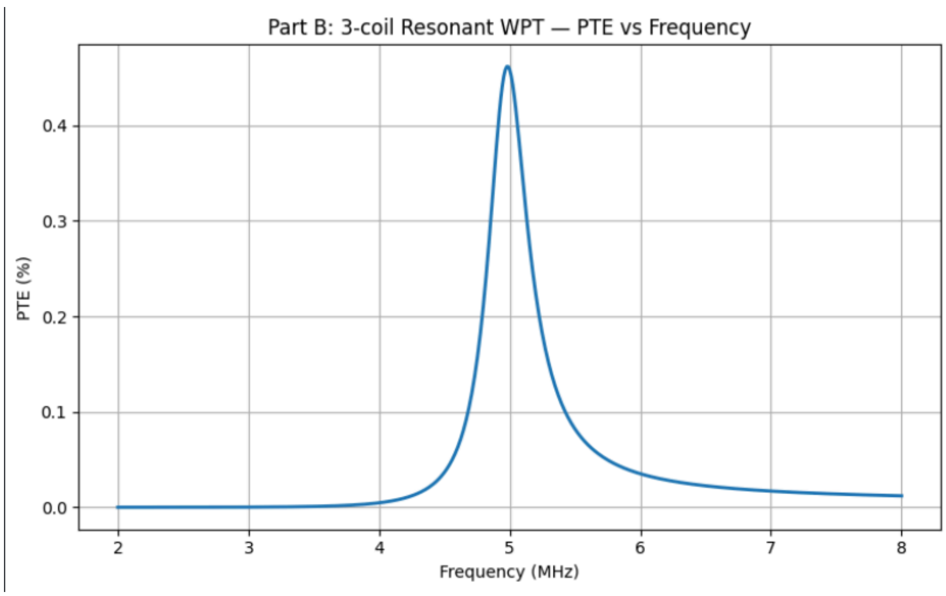

Fig. 8. 3-coil Resonant WPT – PTE vs Frequency

At 5 MHz, the resonant three-coil system achieves PTE = 0.458284%, compared with only 0.001556% for the non-resonant case—a factor of 294.5×. Table II summarises these results. The resonant system exhibits a well-defined peak near 4.978 MHz (maximum PTE = 0.4617%), confirming effective tuning of all three resonators. Compared with the two-coil resonant result, the three-coil resonant architecture offers a 74.4× improvement in PTE at 5 MHz.

Parameter	Non-Resonant	Resonant
PTE at 5 MHz	0.001556 %	0.458284 %
Max. PTE	0.002608 % @ 8 MHz	0.461656 % @ 4.978 MHz
P_Load at 5 MHz	2.576×10^{-8} W	9.111×10^{-5} W
V_Load at 5 MHz	1.135×10^{-3} V	6.749×10^{-2} V

Improvement over non-resonant	—	294.5×

Table II. Three-coil WPT simulation results at 5 MHz.

The improvement arises from two concerted effects. First, resonating all three coils simultaneously maximises current in each coil for a given excitation. Second, the tightly coupled implanted pair (k23 = 0.3) efficiently harvests the energy delivered to the receiver coil and routes it to the load. The sharp resonance peak visible in Fig. 4 is characteristic of a properly tuned cascaded resonator, and its proximity to the design frequency of 5 MHz validates the capacitor selection.

VI. PART C — PARAMETRIC STUDY: PTE VS. DISTANCE

To quantify the dependence of PTE on coil separation, k12 was mapped to physical distance using experimentally motivated values spanning 0–40 mm at 5 mm intervals (Table III). The three-coil k23 was kept fixed at 0.3, reflecting the assumption that the receiver and load coils maintain constant proximity inside the implanted device. LTspice .meas directives extracted current magnitudes at exactly 5 MHz for each distance step.

Distance (mm)	k_{12}	2-Coil PTE (%)	3-Coil PTE (%)
0	0.0350	0.075450	5.343187
5	0.0311	0.059582	4.267218
10	0.0229	0.032313	2.360206
15	0.0153	0.014427	1.067659
20	0.0100	0.006163	0.458925
25	0.0066	0.002685	0.200432
30	0.0045	0.001248	0.093277
35	0.0031	0.000592	0.044288
40	0.0023	0.000326	0.024384

Table III. Distance-coupling lookup and simulated PTE at 5 MHz for resonant systems.

Fig. 5 and Fig. 6 show the PTE versus distance curves for the two-coil and three-coil resonant systems, respectively. Both topologies exhibit a steep monotonic decline: the two-coil PTE falls from 0.075% at contact to 0.00033% at 40 mm (a 230× reduction), while the three-coil PTE falls from 5.34% to 0.024% over the same range. Across every distance, the three-coil system outperforms the two-coil by ~74×, confirming the structural advantage of the three-coil topology. Notably, the 20 mm three-coil PTE (0.459%) is consistent with the Part B base-case result (0.458%) within numerical precision, providing internal cross-validation of the parametric study.

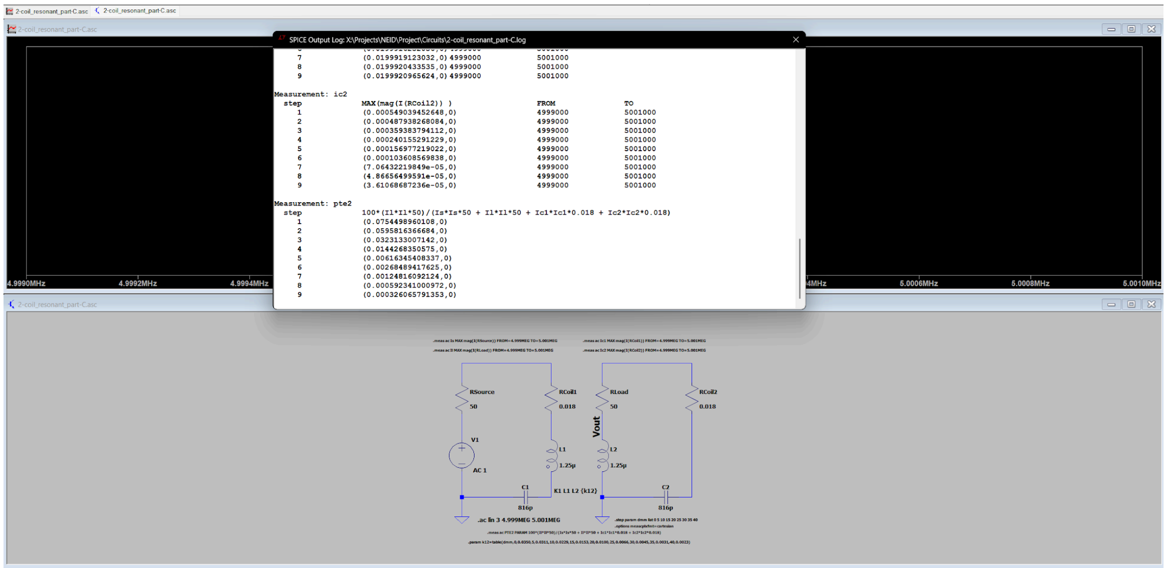

Fig. 9. Simulated PTE vs. coil separation for the two-coil resonant WPT system at 5 MHz.

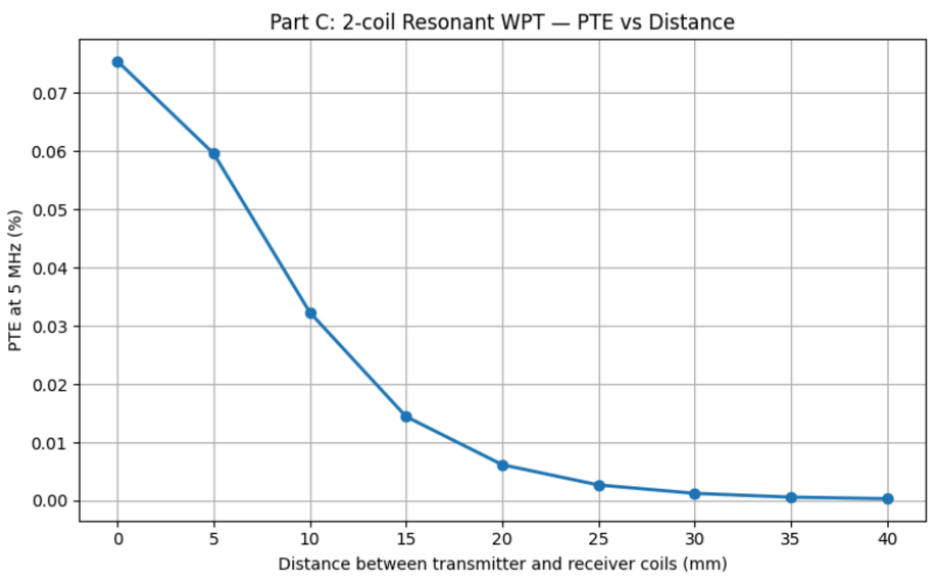

Fig. 10. 2 - coil resonant WPT – PTE vs Distance

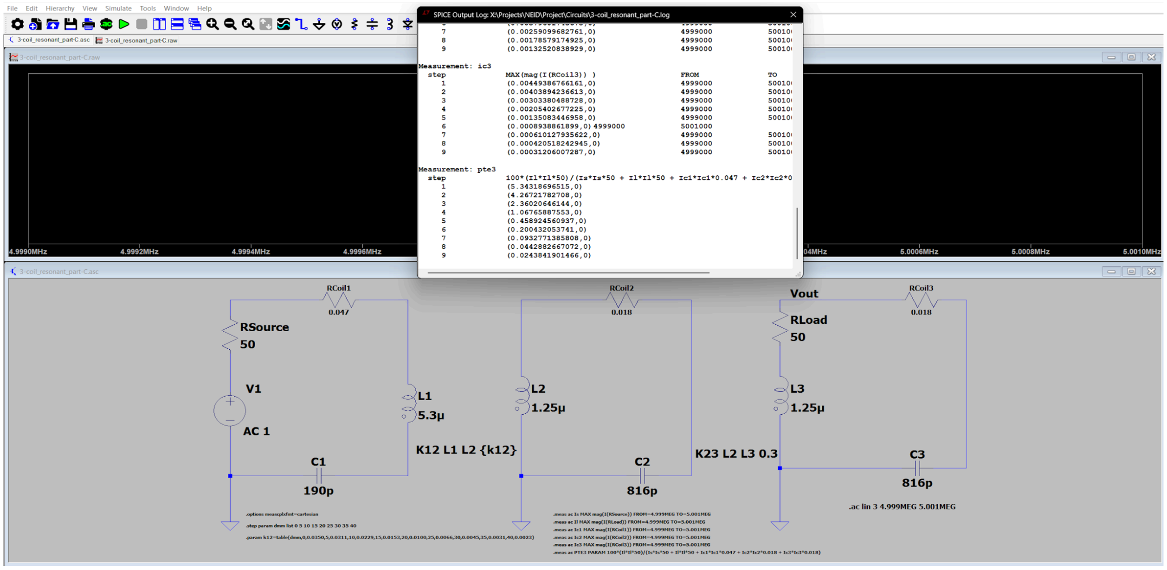

Fig. 11. Simulated PTE vs. coil separation for the three-coil resonant WPT system at 5 MHz.

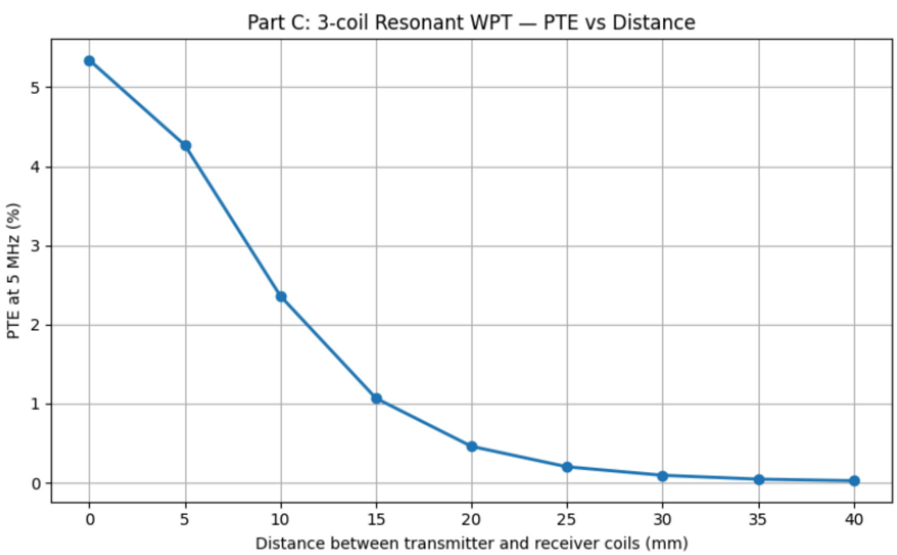

Fig. 12. 3-coil resonant WPT – PTE vs Distance

VII. EXPERIMENTAL VALIDATION

VNA measurements were conducted on the two-coil resonant system at four coil separations: 10, 20, 30, and 40 mm. Fig. 7–10 show the VNA screenshots at each gap. The measured S21 at 5 MHz and the corresponding experimental PTEs computed via equation (8) are collected in Table IV.

Fig. 7. VNA screenshot: S21 measurement at 10 mm gap.

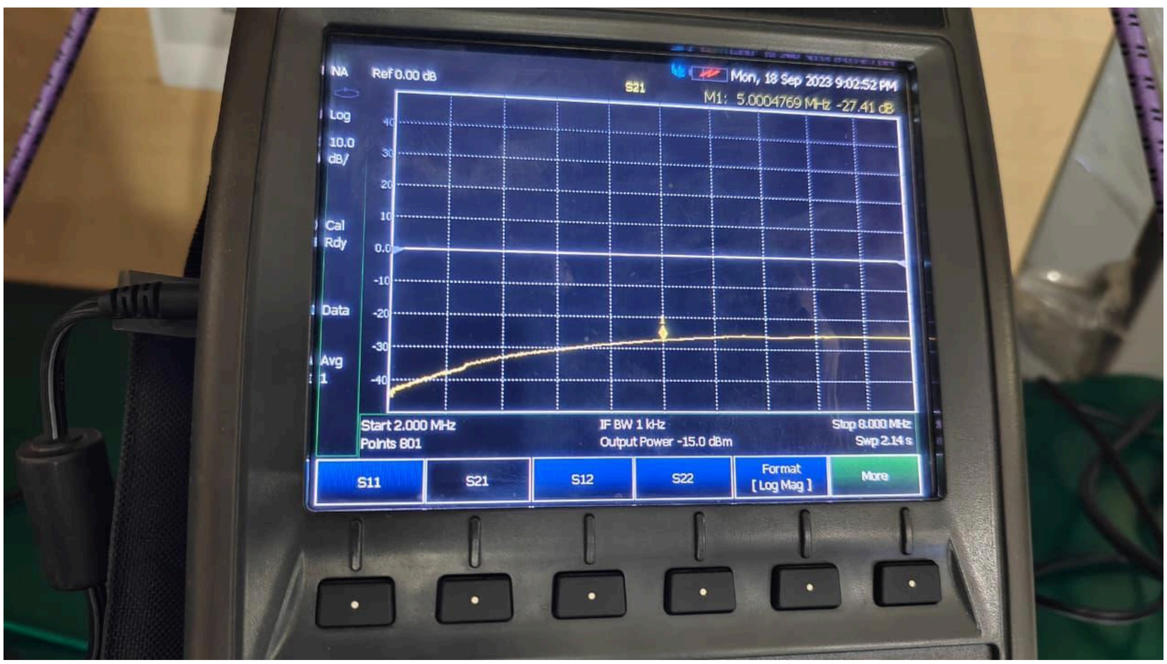

Fig. 8. VNA screenshot: S21 measurement at 20 mm gap.

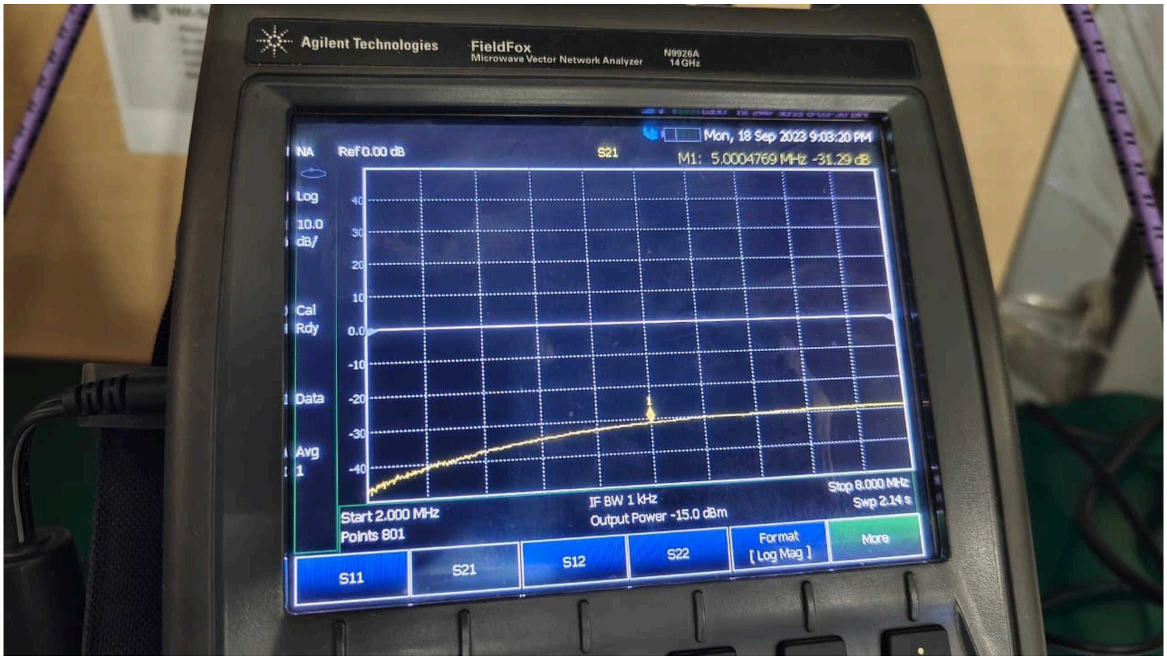

Fig. 9. VNA screenshot: S21 measurement at 30 mm gap.

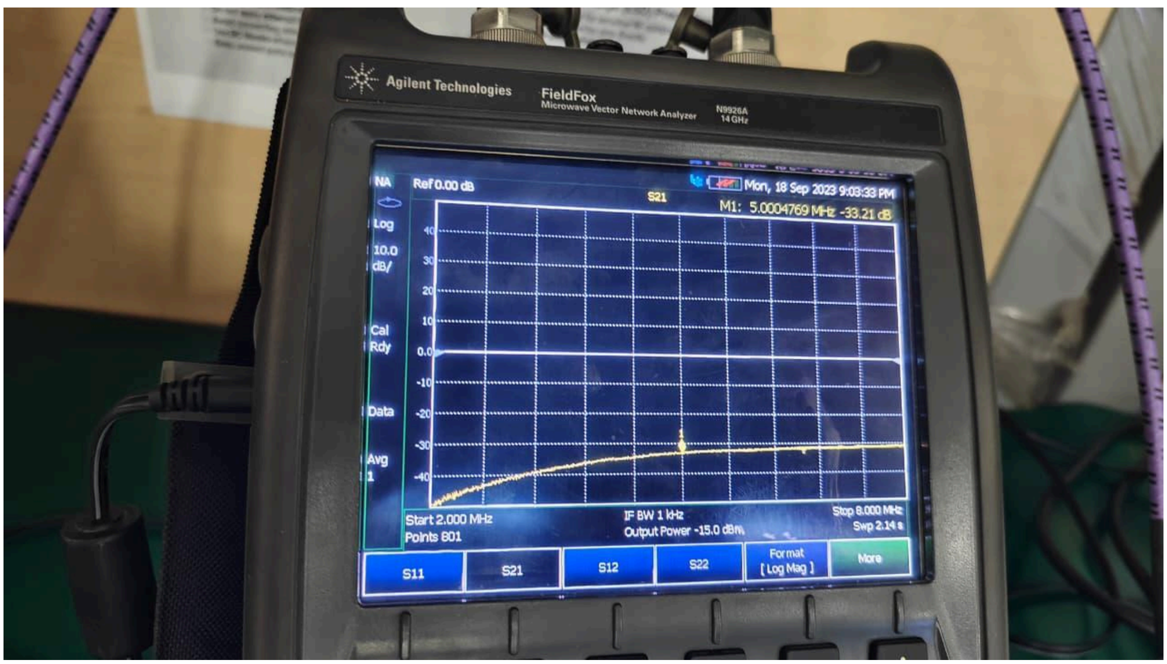

Fig. 10. VNA screenshot: S21 measurement at 40 mm gap.

Gap (mm)	S_{21} at 5 MHz (dB)	Experimental PTE (%)	Simulated PTE (Part C, %)
10	−18.61	1.3772	0.032313
20	−27.41	0.1816	0.006163
30	−31.29	0.0743	0.001248
40	−33.21	0.0478	0.000326

Table IV. Experimental S21 measurements and derived PTE values compared with Part C simulation.

Both experimental and simulated results show a consistent decline in power transfer as the coil gap widens—a qualitative agreement confirming the expected physical trend. However, the experimental PTE values exceed simulated values by factors of 29–146×. This quantitative discrepancy is discussed in Section VIII.

VIII. DISCUSSION

The results across all four parts illuminate several fundamental aspects of resonant WPT design for biomedical applications.

Resonance is indispensable. The factor-of-1.6× PTE gain in the two-coil case and the 295× gain in the three-coil case underscore that non-resonant coupling at 5 MHz is highly inefficient. The impedance mismatch in the non-resonant circuit severely limits current in the coils, directly reducing the mutually induced EMF and the transferred power. Resonance eliminates this mismatch and allows the coils to operate at their maximum current capacity for a given drive level.

The three-coil topology provides transformative performance gains. The tight intra-implant coupling k23 = 0.3 effectively bootstraps the weak transcutaneous link k12 = 0.01. The receiver coil (coil 2) acts as a resonant amplifier, storing energy over multiple AC cycles and delivering it efficiently to the load coil. This cascaded resonance principle is well established in the coupled-resonator literature and is strongly confirmed by the simulation results here.

Distance sensitivity imposes a practical design constraint. PTE scales approximately with $k12^2$, and k12 falls with distance roughly as an inverse power law. Consequently, PTE degrades steeply—by a factor of ~230 over 40 mm even in the three-coil case. For a practical cochlear or retinal implant with 10–20 mm skin depth, this mandates careful transmitter positioning and, ideally, active alignment or feedback control.

Regarding the experimental-simulation discrepancy, several factors jointly explain why measured PTEs are substantially higher than simulated values. The LTspice coupling coefficients are derived from a simplified distance model that may underestimate the true magnetic coupling of the physical coils, which have finite and non-trivial geometry. Additionally, VNA S21 measures the power reaching port 2 relative to power incident on port 1, which includes reflections; this differs from the circuit PTE definition that accounts for source resistance losses explicitly. Real coils also present distributed resonances and parasitic interwinding capacitances that can enhance near-field coupling compared with a lumped-element model. Cable and connector effects, while partially mitigated by calibration, may introduce systematic offsets. Finally, the coils were hand-wound, introducing geometric variability that can increase effective k values. These factors collectively explain a consistent upward bias in experimental PTE relative to simulation.

IX. LIMITATIONS

The principal limitations of this study fall into three categories. First, the LTspice models employ ideal lumped-element components. Real inductors are distributed structures exhibiting skin-effect resistance, interwinding capacitance, and frequency-dependent loss; these are absent from the simulations. The coil resistance values used (0.018–0.047 Ω) are DC approximations; at 5 MHz the skin effect increases effective resistance, which would reduce simulated PTE if included.

Second, the coupling coefficient map (Table III) is based on an empirical approximation rather than direct measurement of k12 at each physical separation. Errors in this map directly propagate to PTE prediction errors, explaining much of the simulation–experiment discrepancy observed in Part D.

Third, the experimental setup introduces several uncontrolled variables: coil planarity and concentricity, cable routing effects on near-field coupling, VNA port-matching accuracy, and environmental electromagnetic interference. Without a controlled fixture ensuring repeatable coil alignment, measurement-to-measurement variability is inherent. Additionally, the S21-to-PTE conversion in equation (8) assumes a perfectly matched 50 Ω two-port, which is approximately but not exactly satisfied when the coil impedances deviate from 50 Ω at off-resonance frequencies.

X. CONCLUSION

This study has demonstrated the design, simulation, and experimental characterisation of two-coil and three-coil inductively coupled WPT systems operating at 5 MHz, targeting biomedical implant applications. Five principal findings emerge.

First, resonant capacitive tuning improves two-coil PTE by 1.6× at 5 MHz, confirming that impedance matching via resonance is essential even for simple inductively coupled systems.

Second, the three-coil resonant topology dramatically outperforms the two-coil resonant configuration, achieving a 295× improvement over the non-resonant three-coil case and a 74× improvement over the two-coil resonant case at the reference operating point ($k12 = 0.01$).

Third, PTE falls steeply with coil separation for both topologies, emphasising the importance of minimising the transmitter-to-implant distance and maintaining coil alignment in clinical deployments.

Fourth, VNA experiments confirm the qualitative trend—S21 becomes more negative (less power transmitted) as the gap increases—providing experimental support for the simulation framework.

Fifth, absolute experimental PTE values exceed simulated values by 29–146×, highlighting the limitations of lumped-element modelling and empirical coupling coefficient estimation. Future work should incorporate frequency-dependent coil models, direct k measurement, and a controlled coil fixture to improve quantitative agreement.

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